CAN Interface Configuration Notes

This document serves as a persistent record of the SocketCAN automation setup for the Selena robot hardware interface. This prevents the "magic" of the automatic connection from being lost during future development.

1. Location of the Configuration

The automation is handled via a **udev rule**. This rule triggers as soon as the PEAK-System USB-to-CAN adapter is plugged in.

File path: `/etc/udev/rules.d/80-can.rules`

2. Current Configuration Rule

The following line is what initializes the `can0` interface at the required bitrate:

```
SUBSYSTEM=="net", KERNEL=="can*", ACTION=="add", RUN+="/sbin/ip link set %k up type can bitrate 500000"
```

3. How to Edit the Rule

To modify settings (like increasing the bitrate to 1,000,000 for 4 motors), open the file using **gedit** with root privileges:

```
sudo gedit /etc/udev/rules.d/80-can.rules
```

4. Applying Changes

After saving the file in gedit, you must reload the udev rules for the changes to take effect immediately:

```
sudo udevadm control --reload-rules
sudo udevadm trigger
```

5. Scaling to Full Gantry (4 Motors)

Optimization Tip: When moving from one motor to the full 4-motor agricultural gantry, the CAN bus traffic will increase. If you experience dropped frames or latency, update the rule to include a larger transmit queue:

```
...type can bitrate 500000 txqueuelen 1000
```

6. Hardware Verification

To verify the interface state and details manually, use:

```
ip -details link show can0
```

Expected adapter type: **pcan_usb** (PEAK-System).